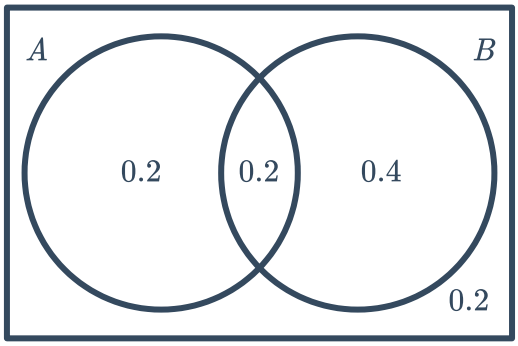
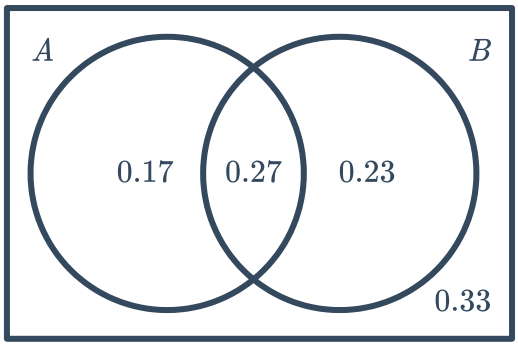
Conditional probability

1 Find $P(A|B)$ given the following probability Venn Diagram:



2 Consider the following probability Venn Diagram. Find:



a $P(A' \cap B)$

b $P(A' \cup B')$

c $P(A|B)$

d $P(B|A')$

e $P(A \cup B|B')$

3 A and B are two random events with the following probabilities:
Find the value(s) of x if:

- $P(A) = 0.3 + x$
- $P(B) = 0.2 + x$
- $P(A \cap B) = x$

a A and B are mutually exclusive.

b A and B are independent.

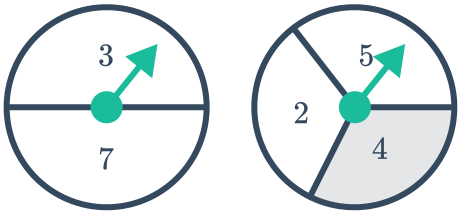
c $P(A|B) = 0.6$

4 The following two spinners are spun and the sum of their respective spins is recorded:

a Construct a table to represent all possible outcomes.

b Find probability that a 7 was spun given that the sum is less than 12.

c Find the probability that the sum is odd given that a 4 was spun.



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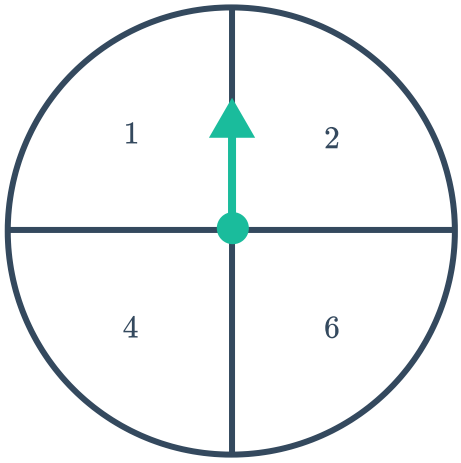
5 The following spinner is spun and a normal six-sided die is rolled. The product of their respective results is recorded.

a Construct a table to represent all the possible outcomes.

b Find the probability that the product was a multiple of 4 given that a 2 was spun.

c Find the probability that a 6 was rolled given that the product was greater than 10.

d Find the probability that the product was greater than 4 given that the same number appeared on the dice and the spinner.



Dependent and independent events

6 Determine whether the following events of selection are independent or dependent.

a A card is randomly selected from a normal deck of cards, and then returned to the deck. The deck is shuffled and another card is selected.

b The selections of each ball in a lottery.

c Two cards are randomly selected from a normal deck of cards without replacement.

d Each student is allowed to randomly pick an item from the teacher's prize bag.

7 Two events A and B are such that $P(A \cap B) = 0.02$ and $P(A) = 0.2$.

a Are the events A and B are independent if $P(B) = 0.1$?

b Are the events A and B are mutually exclusive?

8 Two events A and B are such that $P(A \cap B) = 0.1$ and $P(A) = 0.5$. Calculate $P(B)$ if events A and B are independent.

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9 For each of the following groups of probabilities:

i Find $P(A \cap B')$.

ii Find $P(B \cap A')$.

iii Hence, find $P(A \cap B)$.

iv State whether the events A and B are mutually exclusive.

a $P(A \cup B) = 0.6$, $P(A') = 0.6$ and $P(B') = 0.7$

b $P(A \cup B) = 0.3$, $P(A') = 0.8$ and $P(B') = 0.9$

c $P(A' \cap B') = 0.1$, $P(A) = 0.8$ and $P(B) = 0.3$

d $P(A' \cap B') = 0.2$, $P(A) = 0.1$ and $P(B) = 0.7$

10 For each of the following groups of probabilities:

i Find $P(A \cap B')$

ii Find $P(B \cap A')$

iii Hence, find $P(A \cap B)$

iv Find $P(A) \times P(B)$

v State whether the events A and B are independent.

a $P(A \cup B) = 0.72$, $P(A) = 0.6$ and $P(B) = 0.3$

b $P(A \cup B) = 0.25$, $P(A) = 0.2$ and $P(B) = 0.1$

11 Consider $P(A|B) = 0.8$ and $P(A \cup B)' = 0.15$, where A and B are independent:

a Find $P(A)$.

b Find $P(A' \cap B)$.

c Given that $P(A \cap B) = x$, write an expression for $P(B)$ in terms of x.

d Find the value of x.

e Hence, state the value of $P(B)$.

12 Consider $P(A) = 0.2$ and $P(B) = 0.5$:

a Find the maximum possible value of $P(A \cup B)$.

b Hence, are events A and B mutually exclusive?

c Find the minimum possible value of $P(A \cup B)$.

13 There are 4 green counters and 8 purple counters in a bag. Find the probability of choosing a green counter, not replacing it, then choosing a purple counter.

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14 In a game of Blackjack, a player is dealt a hand of two cards from the same standard deck. Find the probability that the hand dealt:

a Is a Blackjack. (An Ace paired with 10, Jack, Queen or King.)

b Has a value of 20. (Jack, Queen and King are all worth 10. An Ace is worth 1 or 11.)

15 In a game of Draw Poker, a player is dealt a hand of 5 cards from the same deck.

a Find the probability of being dealt a:

i Flush (five cards of the same suit)

ii Royal flush (10, Jack, Queen, King and Ace of the same suit)

b How many possible hands are there?

16 In a game of monopoly, two dice are rolled.

a Is the outcome of the second die dependent on the outcome of the first?

b If a 5 is rolled on one of the dice, find the probability of rolling a 5 on the second die.

17 In a lottery there are 45 balls.

a Find the probability of a particular ball being drawn first.

b A ball is discarded after it has been drawn. If ball number 35 is drawn on the first go, find the probability of ball number 20 being drawn next.

c Is the probability of each successive ball drawn the same as the probability of the first ball drawn?

d Are the draws dependent or independent events?

18 A standard deck of cards is shuffled and placed face down. Dylan attempts to draw five cards randomly so as to get a royal flush (ten, jack, queen, king, ace all of the same suit) .

a If he draws a queen of spades on the first go, find the probability that he draws a ten of spades on the second go.

b If he draws a queen of spades on the first go and a ten of spades on the second go, find the probability that he draws a jack of spades on the third go.

c Is the probability of each draw dependent or independent of the previous draw?

19 A number game uses a basket with 6 balls, all labelled with numbers from 1 to 6. 2 balls are drawn at random. Find the probability that the ball labelled 3 is picked once if the balls are drawn:

a With replacement.

b Without replacement.

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20 A standard deck of cards is used and 3 cards are drawn out. Find the probability, in fraction form, of the following:

- **a** All 3 cards are diamonds if the cards are drawn with replacement.
- **b** All 3 cards are diamonds if the cards are drawn without replacement.

21 A number game uses a basket with 9 balls, all labelled with numbers from 1 to 9. 3 balls are drawn at random, without replacement. Find the probability that:

- **a** The ball labelled 3 is picked.
- **b** The ball labelled 3 is picked and the ball labelled 6 is also picked.

Tree diagrams and probability trees

22 A container holds four counters coloured red, blue, green and yellow. Construct a tree diagram representing all possible outcomes when two draws are done, and the first counter is:

- **a** Replaced before the second draw.
- **b** Not replaced before the next draw.

23 A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.

- **a** Construct a probability tree of this situation with the correct probability on each branch.
- **b** Find the probability of selecting two hearts.

24 Two marbles are randomly drawn without replacement from a bag containing 1 blue, 2 red and 3 yellow marbles.

- **a** Construct a tree diagram to show the sample space.
- **b** Find the probability of drawing the following:
  - **i** A blue marble and a yellow marble, in that order.
  - **ii** A red marble and a blue marble, in that order.
  - **iii** 2 red marbles.
  - **iv** No yellow marbles.
  - **v** 2 blue marbles.
  - **vi** A yellow marble and a red marble, in that order.
  - **vii** A yellow and a red marble, in any order.

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25 Each year at Sicily High School, students take part in a competition to recite the digits in π. Maria won in 2014 and 2015, but there's a new contender, Dario.

a If only these two compete, construct a tree diagram of the possible winners from 2014 to 2018 inclusive.

b If Maria and Dario are equally likely to win each year, find the probability that Dario wins in 2016 given that Maria won in 2018.

c Find the probability that Dario wins two of the three competitions from 2016 to 2018 given that Maria won in 2017.

26 When Fred gets ready for work in summer, he first decides whether it will be a hot day (H) or not (N) and then decides whether to wear a tie (T) or to just dress casually (C).
The chance of Fred deciding it will be a hot day is 0.7. If he decides it will be a hot day, there is a 0.2 chance he will wear a tie. If he decides it will not be a hot day, there is a 0.85 chance he will wear a tie.

a Construct a probability tree for this situation.

b Calculate the probability Fred decided it was a hot day given that he wears a tie. Round your answer to two decimal places.

27 In tennis if the first serve is a fault (out or in the net), the player takes a second serve. A player serves with the following probabilities:

- First serve in: 0.55
- Second serve in: 0.81

a Construct a probability tree showing the probability of the first two serves either being in or a fault.

b Find the probability that the player needs to make a second serve.

c Find the probability that the player makes a double fault (both serves are a fault).

28 Han draws two cards, without replacement, from a set of cards numbered 1 to 4 to create a 2 -digit number. If he draws a 1 then a 2 the number 12 is formed.

a Construct a tree diagram to find all possible outcomes.

b Find the probability that the number formed is:

i Odd

ii Divisible by 11.

iii Even

iv Divisible by 3.

v Divisible by 5.

vi Divisible by 7.

vii Divisible by 4.

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29 Nine pilots from StarJet and 7 pilots from AirTiger offer to take part in a rescue operation. If 2 pilots are selected at random:

a Construct a probability tree showing all possible combinations of airlines from which the pilots are selected.

b Find the probability that the two pilots selected are from:

i The same airline

ii Different airlines

30 Sophia has three races to swim at her school swimming carnival. The probability that she wins a particular race is dependent on whether she won the previous races, as summarised below:

- The chance she wins the first race is 0.7.
- If she wins the first race the chance of winning the second is 0.8.
- If she loses the first race then the chance of winning the second is 0.4.
- If she wins the first two then the chance of winning the third race is 0.9.
- If she lost the first two then her chance of losing the third race is 0.9.
- If she won only one of the first two races, then the chance of winning the third is 0.6.

a Construct a probability tree to represent all outcomes in this situation.

b Calculate the probability Sophia won all three races correct to three decimal places.

c Calculate the probability Sophia won the third race, correct to three decimal places.

Applications

31 Roald has shuffled a standard pack of 52 playing cards. He draws one card from the pack at a time, and then puts it aside. He will stop when he draws the 8 of Clubs.

a Find the probability that he only has to draw one card.

b Find the probability that he has to draw two cards.

c Find the probability that he has to draw three cards.

d Find the probability Roald will stop after the twelth card.

e As he draws more and more cards, does the probability of the next card being the 8 of Clubs get higher, lower or stay the same?

32 Vanessa has 55 songs in a playlist. She starts to play them on shuffle so each song will be played once until all songs have been played.

a Find the probability of Song A coming on first.

b Find the probability of Song A coming on last.

c Is the probability of a particular song playing dependent or independent of the position it plays in?

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33 In a memory game, 16 pairs of identical cards are randomly placed face down. When someone has a go, they turn one card over and then turn a second card over to try to find an identical pair. If an identical pair are found, they are removed from the pack.

- **a** At the beginning of the game, Kathleen turns a card over. Find the probability that she will pick a card that will make an identical pair.
- **b** The first two cards Kathleen picks are identical. Is the probability of picking any more pairs of identical cards independent or dependent on the number of pairs picked before?

34 Vanessa has 12 songs in a playlist. 4 of the songs are her favourite. She selects shuffle and the songs start playing in random order. Shuffle ensures that each song is played once only until all songs in the playlist have been played. Find the probability that:

- **a** The first song is one of her favourites.
- **b** Two of her favourite songs are the first to be played.
- **c** At least one of her favourite songs is played in the first three.

35 Lucy has a box of Favourites chocolates. In this box there are 30 chocolates, 5 of which are Picnics. Lucy takes and eats a chocolate without looking until she gets a Picnic.

- **a** Find the probability she only eats one chocolate.
- **b** Find the probability she eats only two chocolates.
- **c** Find the probability she eats five chocolates.
- **d** As she eats more and more chocolates, is the probability of the next chocolate being a Picnic getting higher, lower or the same?
- **e** How many chocolates must Lucy have eaten to be certain that the next chocolate will be a Picnic?

36 Han is getting a new mobile phone number and he can choose the last 3 digits. He chooses the digits 7, 4 and 6.

- **a** List all the possible numbers Han can make with these three digits.
- **b** If the digits are randomly arranged, find the probability the combination starts with 7 given that it ended with 4.
- **c** Given that the second number is 4, find the probability it ends with 7.

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37 Christa bought four food items, shown in the list, to eat for lunch and dinner - one item per meal. She randomly selects two items for today's lunch and dinner:

- Sandwich
- Soup
- Pasta
- Salad

a Construct a tree diagram to show all her possible choices.

b Find the probability that:

i Soup is selected for lunch and pasta is selected for dinner.

ii Soup and pasta are selected for today's meals.

iii Salad is not selected today.

iv A sandwich is one of the meals selected.

v Salad and a sandwich are chosen for lunch and dinner respectively.

38 Two sets of numbers, A and B, are such that set A contains the even numbers from 1 to 20, inclusive, and set B contains the factors of 48 from 1 to 20, inclusive.

a List the numbers in set A.

b List the numbers in set B.

c Find the total number of unique numbers across both sets.

d Find the probability that a randomly selected number is in set B, given that it is in set A.

39 At an Italian restaurant, Alessandra orders an entree, main and dessert. She has 2 entrees, 3 mains and 2 desserts to choose from:

- **Entrees**: calamari, soup
- **Mains**: spaghetti, lasagna, risotto
- **Dessert**: gelato, tiramisu

a Construct a tree diagram to illustrate the possible combination of meals that Alessandra could order. Alessandra will order her entree first, then her mains, followed by her dessert.

b If each combination of dishes is equally likely, find the probability Alessandra orders lasagna given that she ordered calamari.

c Find the probability Alessandra ordered spaghetti and tiramisu given that she ordered soup.

d Find the probability Alessandra ordered gelato or calamari given that she ordered risotto.

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40 Neil watches three episodes of TV each night. He begins with News or Current Affairs, then the next two shows are either Comedy, Horror or Animation. He never watches more than one Horror show, and if he watched the News, he will follow this immediately with a Comedy.

a Construct a tree diagram of all possible options.

b If, at every stage, the possible outcomes of each choice are equally likely, find the probability Neil watches two Comedies, given that he watched the News.

41 In a survey, 200 people were questioned about whether they voted for Labor, Liberal or Greens last election and who they'll vote for next election. The rows show who people voted for last election and the columns shows who they'll vote for next election:

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	26	41		70
Liberal (last election)			1	90
Greens (last election)	9			
Total		140	10	200

a Complete the table.

b Find the probability that a randomly selected person will vote Labor next election given that they voted Liberal last election.

c Find the probability that a randomly selected person did not vote Green last election given that they will vote Green next election.

d If a person voted Labor or Liberal last election, find the probability they'll vote Liberal next election.

42 For breakfast each morning, Sandy eats porridge, toast or cereal. With that she will either drink orange juice, tea or hot chocolate. Find the probability that Sandy will:

a Eat toast and drink tea.

b Eat cereal or drink orange juice.

c Eat porridge given that she drinks hot chocolate.

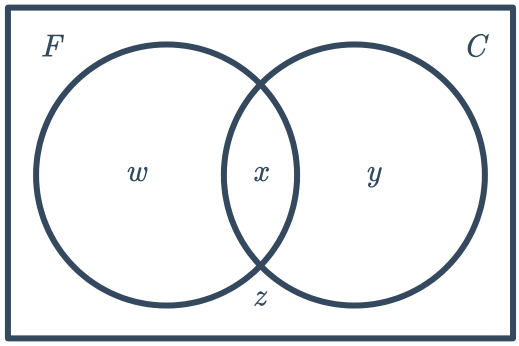
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43 At a university there are 816 students studying first year engineering, 497 of whom are female (set F). Of the 348 students studying Civil Engineering (set C), 237 of them are women.

a Find the value of:

i w **ii** x **iii** y **iv** z

b Find the probability that a randomly selected male student does not study Civil Engineering.



44 In a group of 69 people, 45 said they had been to Europe (E), 26 said they had been to Asia (A) and 8 people said they had been to neither Europe nor Asia. Find:

a $P(A \cap E')$ **b** $n(E \cap A)$ **c** $P(E|A)$ **d** $P(A|E')$

45 In a survey, 87 people are questioned about whether they own a tablet (T) or a smartphone (S). The following probabilities were determined from the results:

$$P(T|S) = \frac{5}{12}, \quad P(S \cap T') = \frac{35}{87}, \quad P(T) = \frac{14}{29}$$

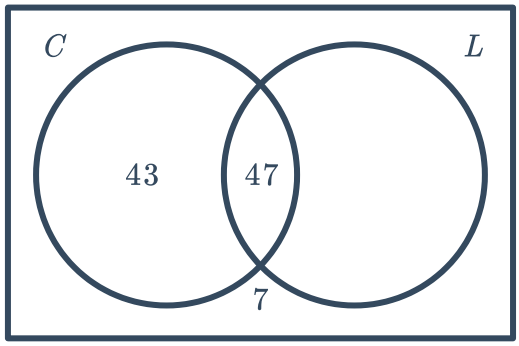
a Find $n(S \cap T)$. **b** Find $P(S' \cap T)$. **c** Find $P(S|T)$. **d** Find $P(T|S')$.

46 In a survey, 100 office workers are asked about whether they buy a coffee (set C) or lunch (set L) during the work day. Some of the results are shown in the given Venn diagram. Find:

a $n(L \cap C')$

b $P(C|L)$

c $P(L|C')$



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47 In a survey, 166 children were asked whether they had read or watched at least one of the stories in the Harry Potter series:

- 62 hadn't read any of the books
- 39 had read and watched at least one of the stories
- 97 had watched the movie
- 4 had never read or watched any of these

a Complete the following table for these results:

	Read the book	Didn't read the book	Total
Watched the movie	39		97
Didn't watch the movie		4	
Total		62	166

b Find the probability that a randomly selected student had:

i Read the book but hadn't watched the movie.

ii Read the book given that they had watched the movie.

iii Seen the movie given that they hadn't read the book.

48 Of the 100 students starting kindergarten surveyed, 80% believe in Santa, 41% believe in the Tooth Fairy and 31% believe in both.

a Complete the following table:

	Believe in Santa	Do not believe in Santa	Total
Believe in Tooth Fairy			
Do not believe in Tooth Fairy			
Total			100

b Of those who believe in Santa, what percentage also believe in the tooth fairy? Round your answer to two decimal places.

c Of those who believe in either Santa or the Tooth Fairy, what fraction believe in both?

d What proportion of those who believe in the Tooth Fairy do not believe in Santa?

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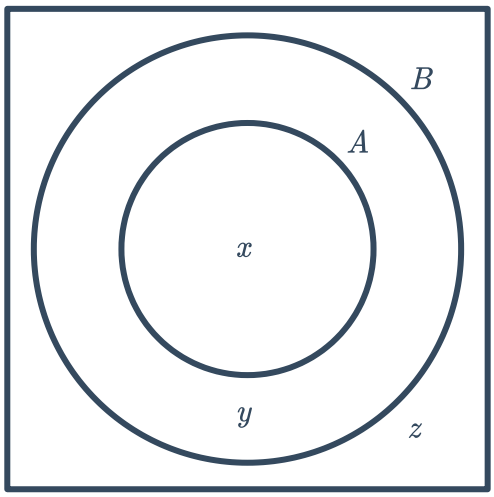
49 Shoppers were surveyed about how much they spend on groceries each week. Of the total surveyed, 426 spent less than $\$200$ each week (set A), 824 spent less than $\$400$ each week (set B) and 154 spent over $\$400$.

a Find the value of:

i x **ii** y **iii** z

b Find the probability a shopper spent less than $\$400$ but more than $\$200$.

c Find the probability a shopper spent over $\$400$ if it is known they spent over $\$200$.



50 In a survey, 200 people were questioned as to whether they read novels in paperback form or on an e-reader:

- 30% said they do both.
- 40% said they read paperbacks.
- The probability that a person who didn't read paperbacks also didn't read on an e-reader was 30%.

a Complete the two-way table for the number of people in each category:

	Paperback	Not Paperback	Total
E-Reader	60		
Not E-Reader			
Total			

b Find the probability that a randomly selected person reads novels on an e-reader but not in paperback.

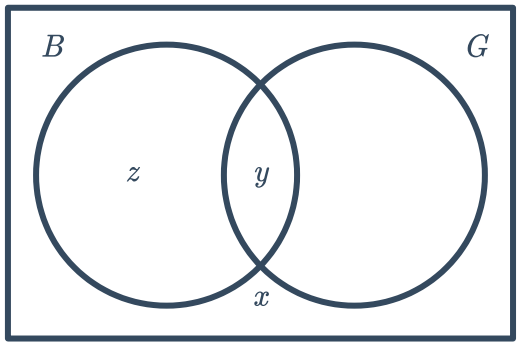
c Of those that read on e-readers, what percentage read in both formats? Round your answer to the nearest percent.

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51 In a survey, 1000 people were asked about whether they buy (set B) or grow (set G) their own herbs. The results are listed below:

- 4% said they do neither.
- 40% said they do both.
- Of those who bought their herbs, the probability that they also grew herbs was $\frac{2}{3}$.

This information is to be displayed in the following Venn diagram, where x, y and z represent numbers of people:



a Find the value of: x and y.

i x **ii** y **iii** z

b Find the number of people who only grew their own herbs.

c Of those who grew herbs, what proportion also bought herbs?

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